

A RECORDER-RELATIVE EVIDENCE AUDIT OF SCHEDULE-ALIGNED IDENTITY-LOCK PERIODICITY

Anonymous authors

Paper under double-blind review

ABSTRACT

Position-decomposed diagnostics of diffusion large language models can confuse sampler geometry with model behavior because an active-block recorder exposes positions only through the current sampling window. We present a recorder-relative evidence audit of one case study that holds the model, questions, canvas, decoding rule, and total function evaluations fixed while applying a composite intervention: block partition changes together with the required per-block step allocation. For the two main arms (L16/L64) and one reused L32 development control, the source-reported complete candidate rows are arithmetically consistent with the selector returning the configured length. The source also reports the same ordering under a rank-sum statistic computed from those profiles; it is neither independent evidence nor calibrated inference. Sequential canvas-96 L24/L48 observations are excluded from the headline result because their complete candidate-separation and rank-sum vectors are absent, so their argmaxes cannot be audited. We retain them only as side observations: the reported L24 conditional margin is marginal and the L48 separation rests on one boundary locus. We recommend reporting separately what boundary-set composition makes reachable before data and what measured magnitudes support. The result is descriptive for the active recorder and supports no model, sampler, recorder-causal, or allocation-causal mechanism.

1 INTRODUCTION

Diffusion language models generate discrete sequences by iterative denoising (Austin et al., 2021; Zheng et al., 2024; Lou et al., 2024; Shi et al., 2024; Nie et al., 2025). Block diffusion provides the closest verified architectural context: it is autoregressive across blocks and applies discrete denoising within each block (Arriola et al., 2025). As decoding schedules become variable or structured (Li et al., 2025; Sun et al., 2026; Wang et al., 2026), a position-resolved diagnostic can inherit periodicity from the schedule that exposes its positions. Without an intervention, such a readout cannot distinguish model-intrinsic state, sampler state, and what the recorder makes observable.

We examine identity-lock time on a blockwise-decoded LLaDA-style model (Nie et al., 2025; 2026). An apparent period of 32 motivates a matched composite intervention over the block partition and its required per-block step allocation: the model, 300 questions, canvas, total NFE, decoding rule, and temperature remain fixed. The active-block recorder, however, logs only masked positions in the currently active block. Its lock-time profile is therefore defined relative to the sampling window (Figure 1).

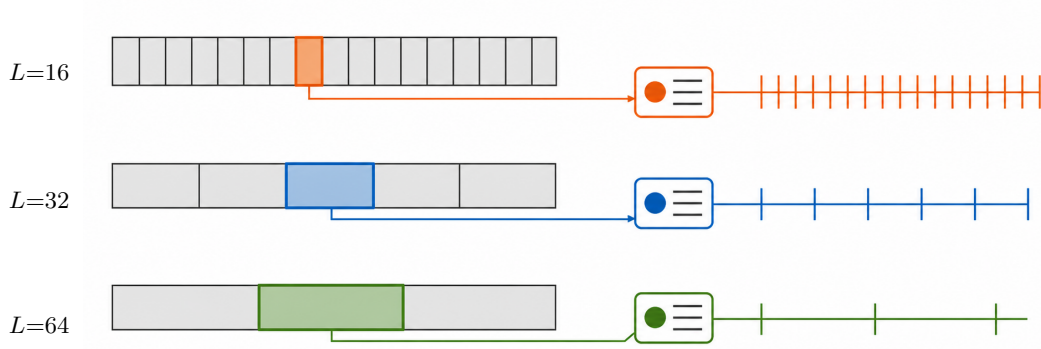


Figure 1: Qualitative measurement pipeline. From top to bottom, block length L grows while canvas extent stays fixed; color marks the one active block exposed to the recorder, and the right-hand ticks depict the boundary spacing visible through that window. Pulse and group counts are schematic, not quantitative evidence. Measured profiles and separations are in Figure 2 and Table 3.

We frame the contribution as a recorder-relative evidence audit of a narrow case study. It makes three scoped contributions. First, the complete printed candidate rows for the two main arms and reused L32 development control are consistent with $P^* = L$; the reported rank-sum ordering is a same-profile, uncalibrated sensitivity statistic rather than independent confirmation. Second, measured candidate-32 signs are retained as descriptive observations at the recorder-output layer. Sequential L24/L48 probe cells are kept only as unauditable side observations, not as headline argmax confirmation. Third, the case study motivates a reporting recommendation: specify before data which outcomes boundary-set composition makes reachable, then report separately what the measured magnitudes show. It does not establish a model, sampler, recorder, allocation, cross-bed, or universal mechanism.

2 MEASUREMENT AND DESIGN

Core notation is defined here before it is used; Appendix C retains the longer terminology crosswalk.

Notation. For decode i and relative canvas position r , $\ell_{i,r}$ is the recorded identity-lock time. Its positionwise median is $m(r)$ and $d(r) = m(r+1) - m(r)$. For candidate period P , $\mathcal{B}_P$ contains candidate boundary indices and $\mathcal{I}_P$ their interior complement; $\text{sep}(P)$ is the boundary-minimum minus the interior-maximum of $d(r)$, and P^* maximizes it. “Canvas 96” denotes the separate 96-position probe bed. $\text{BIN}k$ denotes a restricted content null with k equally populated offset-rank strata.

Evidence status and paper identity. Unless a quantity is explicitly introduced as a mathematical definition or an arithmetic identity, every empirical number, count, schedule, estimate, band, re-sampling summary, duration, and software version in this paper has status SR/CV: *source-reported and cannot be verified locally*. The audited package contains the manuscript and two raster figures but not the event streams, analysis programs, complete registration records, or machine-readable results. No analysis was rerun. Direct authorial language below describes the audited design and source-reported readouts, not a new execution. Complete L24/L48 candidate-separation and rank-sum vectors are absent and are not reconstructed; those probes therefore remain side observations without an auditable argmax. This paragraph supplies the paper-wide provenance rule; later caveats are repeated only where they change the interpretation of a result.

Model bed. We use an LLaDA-style masked dLLM at the LLaDA-8B-Instruct scale with a locally registered checkpoint on GSM8K (Cobbe et al., 2021). Decoding uses temperature 0 and `remasking=low_confidence`, the confidence-ordered parallel decoding rule introduced for conditional masked language models and used by LLaDA (Ghazvininejad et al., 2019; Nie et al., 2025). The main differential fixes NFE at 48 and uses the same 300 questions. Its L32/NFE48 arm is a frozen *development control*: a registered contiguous $n=300$ subset of a pre-existing pool

Element	Main arms	Reused L32 control	Sequential probes	Content bed
Canvas / NFE	256/48 16 : (4, 4, 4, 4, 4, 4, 4, 4, 2, 2, 2, 2, 2, 2, 2, 2) $8 \times 4 + 8 \times 2 = 48$ 64 : (16, 16, 8, 8) 16 + 16 + 8 + 8 = 48	256/48 32 : (8, 8, 8, 8, 4, 4, 4, 4) $4 \times 8 + 4 \times 4 = 48$	96/48 24 : (12, 12, 12, 12) $4 \times 12 = 48$ 48 : (24, 24) $2 \times 24 = 48$	256/32 32; ordered vector not supplied
L ; ordered per-block steps	same 300	contiguous development subset, $n=300$ (opaque IDs)	same 300	GSM8K
Candidate grid Resampling	{8, 16, 32, 64} period $B=2000$; stability/sign $B=1000$	{8, 16, 32, 64} period $B=2000$	{8, 16, 24, 32, 48, 64} period $B=2000$	— interval $B=500$; null $R=2000$

Table 1: Unified configuration (all empirical entries are SR/CV). All beds use the same model family, task, temperature, and remasking rule. The displayed NFE-48 schedules expose every ordered block allocation and its arithmetic total; the content bed’s ordered vector is not available. The control’s contiguous development subset is not part of the non-nested probes. Those two arms form a separate canvas-96 setup with $n=300$, NFE 48, the displayed schedules and grid, and 2,000 cluster-bootstrap replicates. NFE denotes total function evaluations. At fixed total NFE, changing the partition also changes the required per-block step allocation; the design is therefore a composite intervention and does not isolate an allocation effect.

whose internal identifiers are omitted from blind prose. The L32/NFE32 content bed is separate and supports only Section A.7.

Setup and protocol. Table 1 consolidates every bed and schedule. The headline length metric is $d(r) = m(r+1) - m(r)$, the adjacent difference of median identity-lock time. For content, the same lock-time ordering is compared with residualized confidence under an explicit eight-stratum offset-rank null (BIN8; four-position granularity), plus a same-amplitude *position-fake*: confidence values are assigned to positions to preserve amplitude while destroying the observed content-to-position pairing. Intervals use cluster-bootstrap percentiles over units (Davison & Hinkley, 1997); decode resampling preserves multiplicity.

Identity-lock time and finite observation domain. For canvas length C , generated positions lie in $\mathcal{R}_C = \{0, \dots, C-1\}$. For an analyzed decode–position pair (i, r) , let $\mathcal{T}_{i,r}$ be the finite, ordered set of step labels at which the active recorder contains position r ; $x_{i,r,s}$ is the predicted token at $s \in \mathcal{T}_{i,r}$ and $x_{i,r}^f$ its recorded final value. The lock time is

$$\ell_{i,r} = \min\{s \in \mathcal{T}_{i,r} : x_{i,r,t} = x_{i,r}^f \text{ for every } t \in \mathcal{T}_{i,r}, t \geq s\}. \quad (1)$$

Thus, $\ell_{i,r}$ is the earliest recorded step after which the token identity never changes again within that position’s recorded domain. The available package does not establish whether a step label is a global function-evaluation index, a block-local step, or a position-observation ordinal. It also lacks the event-schema rule for empty observation sets, absent final values, censoring, or terminal/unresolved positions. Equation (1) is therefore defined only for source-analyzed pairs with nonempty $\mathcal{T}_{i,r}$ and an available $x_{i,r}^f$; the exact inclusion/exclusion or encoding rule remains unresolved and cannot be audited. Let $\mathcal{A}_r$ denote that unavailable source-defined set of analyzed decodes. The reported profile is $m(r) = \text{median}_{i \in \mathcal{A}_r} \ell_{i,r}$ and $d(r) = m(r+1) - m(r)$. All analysis uses relative generated-canvas position, not prompt position.

Min–max period separation and registered selection criterion. Adjacent differences have the finite index domain $\mathcal{D}_C = \{0, \dots, C-2\}$. For each candidate period in the arm’s ordered registered grid, define $\mathcal{B}_P = \{r \in \mathcal{D}_C : r \bmod P = P-1\}$ and $\mathcal{I}_P = \mathcal{D}_C \setminus \mathcal{B}_P$. A candidate is admissible only when it belongs to that grid and both sets are nonempty. The registered estimator is

$$\text{sep}(P) = \min_{r \in \mathcal{B}_P} d(r) - \max_{r \in \mathcal{I}_P} d(r), \quad P^* = \arg \max_P \text{sep}(P). \quad (2)$$

Exact ties select the first (smallest) period in the ordered registered grid. Positive separation requires every candidate boundary jump to exceed every interior jump. On a strictly L -supported profile it penalizes divisors with false boundaries and multiples that leave true boundaries inside; without that condition it is not a general divisor/multiple guarantee. The registered selection-and-stability criterion fires only under the conjunction $\text{sep}(P^*) > 0$ and a cluster-bootstrap modal period equal to P^* with mode fraction at least 0.95. This operational criterion is not a null-calibrated hypothesis test: its bootstrap mode fraction measures stability under decode resampling, not a false-positive rate. The source protocol reports resampling the 300 decodes with replacement, preserving multiplicity, for $B=2,000$ replicates. Replicate-level outputs needed to reconstruct the criterion’s nesting are not in the audited package.

Selection stability and its honest reading of “no misses.” Two resampling readouts are source-reported. First, a *joint paired cluster bootstrap* resamples a shared index set of the 300 decodes once and applies the same indices to the two main arms, the reused development control, and the two sequential canvas-96 probes. Because the exact pairing manifest and all probe replicate indicators are absent, the headline relation is limited to the two main arms and reused development control. Across $B=2,000$ replicates under a fixed opaque seed, the source reports $\Pr[P_{16}^*=16 \wedge P_{32}^*=32 \wedge P_{64}^*=64] = 1.0000$; it also reports marginal $\Pr[P^*=L] = 1.0$ for each of these three profiles (Table 9). Shared numeric indices alone do not establish semantic pairing. Second, Table 2 reports $\Pr[\text{sep}(L) > 0]$ rather than “0 misses.” The package does not resolve how this positivity indicator was nested with the separately computed modal-period fraction, so the table does not label it as a recomputed conjunction.

Integer lock times do *not* imply integer medians or separations. With an even sample such as $n = 300$, the conventional midpoint median can be half-integral; the reported L24/L48 values contain such half-steps. The exact median implementation is unavailable, so its convention cannot be audited. Under the conventional rule, $m(r)$, $d(r)$, and $\text{sep}(P)$ lie on a half-step lattice. This discreteness permits repeated values, but does not by itself force a bootstrap point mass. The collapsed L16 band is therefore an observed source-reported property of that bootstrap distribution, not a consequence of integer-valued medians. A 0/2,000 miss count is read only as sample-selection stability, never false-positive control.

Because sep is an extremum statistic, we also report a *non-extremal* alternative readout: a rank-sum comparison between candidate-boundary and candidate-interior values of the *same* $d(r)$ profiles. For the two main arms and reused control, the source reports that its maximizing candidate is L . This is a same-profile sensitivity statistic, not independent evidence; its dependence, tie correction, normalization, and inferential calibration are unavailable. The values labeled “ z ” in the source are retained below only as uncalibrated standardized summaries. Complete candidate score vectors are absent, so even this three-profile ordering cannot be audited from the package. A source-reported quantile variant is incomplete and is not used as confirmation of the extremum statistic.

Readout	L32		
	L16	control	L64
$\Pr[\text{sep}(L) > 0]$, joint bootstrap	1.0	0.8805	0.9995
Reported rank-sum “ z ” at L (uncalibrated)	13.9	8.3	4.4
Wrong-candidate fire rate, original three-bed probe	0.0	0.0	0.0

Table 2: Source-reported main-arm/control selection stability and same-profile alternative readout (SR/CV; neither is a false-positive rate). The first row reports only the supplied positivity fraction; replicate-level data needed to reconstruct a nested two-part conjunction are unavailable. The L32 control’s 0.8805 is the weakest arm and appears alongside the source-reported modal-argmax relation. The second row retains the source’s standardized rank-sum label, but is not a calibrated z test and has no independent-evidence interpretation. The final row is a separate $B=1,000$ in-place wrong-candidate diagnostic for the original three beds: its zeros follow from negative point estimates and are not fresh-null false-positive rates. Within an arm, cells share draws. Availability is summarized in Table 9.

Replication accounting. Table 1 is the complete resampling inventory. Period selection and its interval share one draw, whereas the sign and stability checks use separate draws; Table 3 identifies the shared readouts. Uncounted values are observed-data point estimates; artifacts are in Table 9.

Matched intervention and scope. The two treatment arms and frozen development control are in Table 1. At fixed total NFE, changing the partition necessarily changes per-block step allocation, so the treatment is partition-plus-required-allocation. A source-reported within-arm rank transform preserves $P^*=L$ and each candidate-32 sign for the two main arms and reused control (Table 9); a uniform-allocation arm was not registered. The result therefore identifies a recorder-output association with the composite partition-plus-allocation schedule, not which component places or scales the boundary. A factorial allocation/partition intervention is future work.

Registration and two-layer reachability. The audited record states that the estimator and selection-and-stability criterion were fixed before treatment, but amendment v3 withdrew the model-intrinsic decision branch after treatment. The registration files and timestamps are absent, so this chronology cannot be independently audited; Appendix A, Table 8, retains only an opaque sequence. The canvas-96 probes were registered separately before their execution. We therefore report reachability at two layers. *A priori*, nesting such as $B_{32} \subset B_{16}$ determines set composition but not the sign of $\text{sep}(32)$. *From events*, the observed magnitude ordering gives $\text{sep}(32) = -2$ on L16 and a negative value on L64. Thus a recorder-observable fixed-32 component is not selected under these active-block regimes; this does not identify model-intrinsic or sampler-state causes. The exact loci and synthetic condition are consolidated in Table 6; the legacy verdict labels remain only in Appendix C.

Separately conditioned content analysis. On the separate L32/NFE32 bed, BIN8 permutes confidence within eight equally-populated offset-rank bins after a fixed nuisance fit. The initial confidence `conf0` is read at the first recorded step of a position’s block, following confidence-ordered parallel decoding (Ghazvininejad et al., 2019; Nie et al., 2025). This supporting diagnostic is not a cross-block-length validation.

3 RESULTS

3.1 RECORDER-RELATIVE CANVAS-256 RESULT

The audited record reports that the reused development control recovers its known period and the two main arms move to their respective block lengths (Figure 2; Table 3). For these three canvas-256 arms, the printed complete four-candidate rows show that only the selected period has positive min-max separation. These empirical cells are source-reported, not locally rebuilt. The paired main-arm/control relation is reported in all 2,000 shared-cluster resamples. A rank-sum ordering of the same three profiles gives the same reported candidate; it is an alternative statistic, not independent calibrated inference. Table 5 retains the source-reported main-arm/control conditional margins as descriptive diagnostics for one within-block shuffle construction, not significance, rejection, false-positive-rate estimation, or family-wise calibration. Probe `argmaxes` are excluded from this result and from Table 3.

Candidate 32 under the active-block recorder. Table 3 reports $\text{sep}(32) = -2$ on the measured $L = 16$ arm. The exact values in Table 6 show why: candidate-32 boundary values remain positive, but the measured odd-16 interior is larger. The joint sign $\text{sep}(16) > 0 \wedge \text{sep}(32) < 0$ holds in all 1,000 dedicated bootstrap replicates. Candidate 32 is therefore not selected at the recorder-observable layer on this bed. Nesting alone does not force that sign, and the result says nothing about an unobserved model-intrinsic or sampler-state component.

L (source)	$\text{sep}(P)$				P^* (mode frac.)	$\text{sep}(P^*)$ stability band [†]	$(\min_B, \max_I)$		
	$P = 8$	$P = 16$	$P = 32$	$P = 64$			@ P^*	$ B_{P^*} $	$ I_{P^*} $
16 (measured)	-1	+1	-2	-2	16 (1.0)	[+1, +1]	(+2.0, +1.0)	15	240
32 (control)	-1	-1	+1	-1	32 (1.0)	[0, +2]	(+2.0, +1.0)	7	248
64 (measured)	-1	-1	-1	+2	64 (1.0)	[+1, +3]	(+3.0, +1.0)	3	252

Table 3: Source-reported development-control and main-arm results (SR/CV). Each arm is reported with 2,000 cluster-bootstrap replicates; the mode fraction and percentile band are paired readouts of the same draw. The $(\min_B, \max_I)$ columns report the boundary minimum and interior maximum, followed by their observed set cardinalities. The four printed cells in each row are the complete candidate grid. Half-integral values are compatible with an even-sample conventional median. The bands diagnose selection stability rather than coverage; L16 is reported as a point mass. Tables 1, 2, 5, 7, and 10 give the complete resampling inventory and separately budgeted checks; availability is in Table 9.

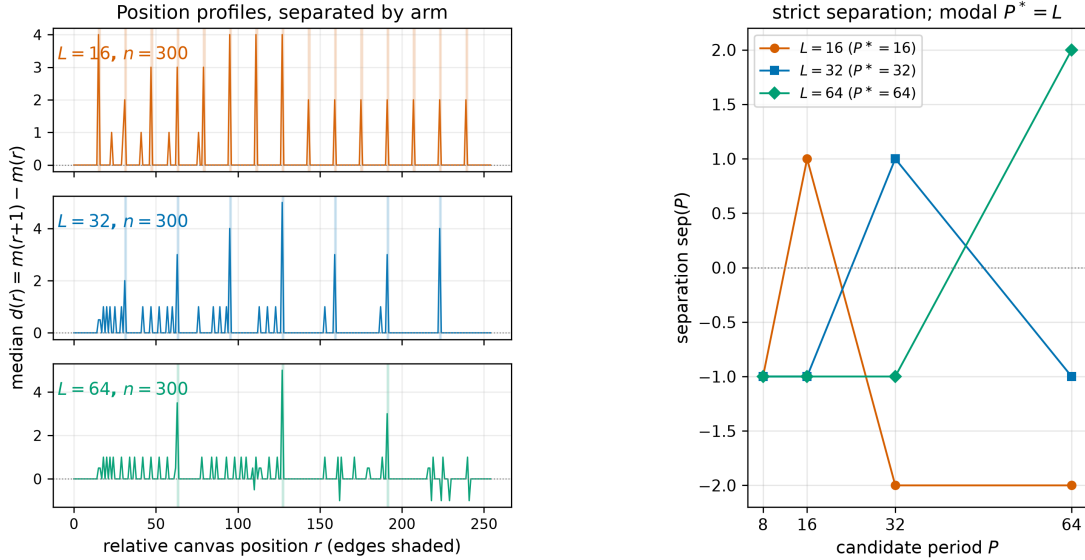


Figure 2: Pre-existing differential raster (all plotted measurements are SR/CV). Left: three stacked facets isolate the median adjacent-difference profile $d(r) = m(r+1) - m(r)$ for L16 (orange), the L32 frozen control (blue), and L64 (green), each from $n=300$ decodes; shaded ticks mark that arm’s block edges. The control is a reused contiguous development subset. Right: only the matching candidate has positive separation in each displayed arm. The source data and generating program are absent, so this figure was not regenerated for the audit and cannot be independently verified. The recorder-level comparison does not distinguish sampler from model mechanism.

Probe arm	Supplied sep(32)	Supplied limitation/readout
$L=24$ probe	-8.0	$P=48$: -1.0; band $[-3.0, 0.0]$; $p_{\text{margin}} = 7.1 \times 10^{-2}$ (marginal) reported selected-boundary count 1;
$L=48$ probe	-7.5	$p_{\text{margin}} = 2.1 \times 10^{-2}$

Table 4: Unauditable side observations for the sequential canvas-96 probes (SR/CV), excluded from the headline result. Candidate 32 is disjoint from the $L=24$ boundaries. The displayed $P=48$ cell is mismatched; the separately source-reported p_{margin} for L24 is marginal. The percentile band is a selection-stability diagnostic, not a coverage-valid interval. These are not complete candidate vectors: the omitted L24/L48 values and all-candidate rank-sum scores are absent, so no probe argmax is claimed or auditable here. The L48 count records the source-reported single-locus limitation, not confirmation.

Arm	P^*	sep(P^*)	m_{obs}	p_{sepL}	p_{margin}
L16 (canvas 256)	16	+1.0	+2.0	5.0×10^{-4}	1.5×10^{-3}
L64 (canvas 256)	64	+2.0	+3.0	5.0×10^{-4}	1.0×10^{-3}
L32 control (canvas 256, NFE 48)	32	+1.0	+2.0	5.0×10^{-4}	1.0×10^{-3}

Table 5: Source-reported main-arm/control conditional separation-margin diagnostic (SR/CV; not full-selector calibration). Each of $R=2,000$ replicates under an opaque fixed seed permutes $d(r)$ within each true L -block, including its edge index, then records the argmax candidate’s top-minus-second gap. Here $m_{\text{obs}} = \text{sep}(P^*) - \max_{c \neq P^*} \text{sep}(c)$ and the discrete exact $p = (1 + \#\{m_{\text{perm}} \geq m_{\text{obs}}\})/(R+1)$. The p_{sepL} column tests $\text{sep}(P^*)$ alone; p_{margin} is the selection-controlled readout. Provenance and the original-gate size audit are in Table 9 and Appendix A.5.

Measured class	Loci	$d(r)$ values	Readout
L32 true-32 boundaries	[31, 63, 95, 127, 159, 191, 223]	[2, 3, 4, 5, 3, 3, 4]	development control
L16 true-16 odd chain	[15, 47, 79, 111, 143, 175, 207, 239]	[4, 3, 3, 4, 2, 2, 2, 2]	candidate-32 interior max 4
L16 would-be $\mathcal{B}_{32}$	[31, 63, 95, 127, 159, 191, 223]	[2, 3, 4, 4, 2, 2, 2]	boundary min 2; sep(32) = -2
L64 mid-block 32	[31, 95, 159, 223]	L64: [0, 0, 0, 0]; L32: [2, 4, 3, 4]	recorder-observable collapse
L64 true-64	[63, 127, 191]	L64: [3.5, 5, 3]; L32: [3, 5, 3]	true boundary remains
Strict-support synthetic L		Source periods	Selected P^*
		16	{8, 16, 32, 48}
		32	{8, 16, 32}
		64	{8, 16, 32}

Table 6: Consolidated negative evidence, reachability, and development check. A priori, nested boundary sets determine membership but do not force the sign. From events, L16 candidate 32 has boundary minimum 2 and interior maximum 4; the joint sign holds in 1,000/1,000 bootstrap replicates. On L64, would-be 32 boundaries are recorder-observable interiors. The lower panel is a development-only synthetic control: $P^* = L$ holds for the shown strictly L -supported profiles, not for arbitrary residuals. The upper rows are observed-data readouts from their arm samples, not additional replications. Provenance is in Table 9.

3.2 SELECTION STABILITY, NOT NULL CALIBRATION OR MECHANISM

The central estimand is discrete period identity. On each supplied profile, positive min–max separation is a point-estimate property; the source-reported bootstrap probes stability to resampling decodes. One shared question-index draw is reported across the two main arms, reused development control, and two sequential probes (Section 2), but the headline uses only the source-reported three-profile main-arm/control relation. Shared indices do not establish semantic pairing. For those three profiles, the reported positivity fractions range from 0.88 to 1.0, with the reused L32 development control weakest at $\Pr[\text{sep}(32) > 0] = 0.8805$. The alternative rank-sum ordering uses the same three profiles and has no independent or calibrated-inference status.

The L32 control reuses a pre-existing contiguous pool and its percentile stability band $[0, +2]$ touches zero (it is not a coverage-valid CI). A fresh in-session L32 control remains unexecuted future work.

Exploratory spectral residual check. Because sep cannot exclude every residual shape for a nested candidate, a post-data DFT check tests one sinusoidal residual-32 family on L16. Its injected-residual control fires and its no-residual control is silent; the real L16 profile does not cross the family-wise threshold at $P = 32$, and a direct boundary-subset test finds no amplitude split. This is a narrow negative for the tested family, not evidence that arbitrary residual shapes are absent. The estimator, complete candidate grid, power sweep, and thresholds are reported in Appendix A.1 and Table 7.

Canvas-96 probe side observations. On the separate canvas-96 beds, candidate 32 is non-nested relative to L24 ($\mathcal{B}_{32} \cap \mathcal{B}_{24} = \emptyset$). Table 4 retains only supplied candidate cells and limitations: L24 is marginal under the source-reported conditional margin, and the reported L48 selected separation has one boundary locus. The complete six-candidate separation and rank-sum vectors are absent, so neither probe winner is asserted or checked in this evidence audit. These sequential observations do not replace the matched canvas-256 arms and support no cross-canvas generalization.

The L64 mid-block reversal and its L32 comparator are reported without a separate verdict vocabulary in the consolidated negative-evidence Table 6.

Supporting analysis outside the headline claim. A separately budgeted L32/NFE32 bed shows a moderate-granularity association between initial confidence and later lock ordering. Because it is not a cross-block-length validation and does not identify a schedule mechanism, its estimand, results, and null-granularity audit are reported in Appendix A.7.

4 LIMITATIONS

- **Observation is not mechanism.** The active-block recorder cannot separate model-intrinsic state, sampler state, and recorder-observable state. Candidate 32 is not selected only at the last layer under the tested regimes.
- **Registration history.** The estimator was selected with a failed development control before treatment, but the model-intrinsic decision branch was withdrawn after treatment. The non-nested probes have their own pre-run registration and a different 96-position canvas.
- **Bed and grid.** Evidence covers one model family, one task, and the candidate grids in Table 1; off-grid periods, other models, tasks, samplers, and untested block lengths are not covered.
- **Composite intervention.** Partition and required per-block step allocation vary together. Boundary location is measured; a distinct allocation effect on amplitude is not.
- **Boundary cardinality.** The min–max separation is an extremum statistic, so its support matters: on the probe arms the selected period rests on few measured boundary positions (the L48 probe has $|B_{48}|=1$), making its positive separation a single-locus readout rather than a stable margin. We retain that source-reported limitation in Table 4; it is not headline confirmation.

- **Probe auditability.** The audited package omits the complete L24/L48 candidate-separation vectors and all-candidate rank-sum scores. The source-reported probe argmaxes cannot be independently audited and are excluded from the headline claim.
- **Event semantics.** The event schema and recorder implementation are absent. The ordering domain $\mathcal{T}_{i,r}$ is finite, but whether its labels are global, block-local, or per-position and how missing, censored, or terminal cases enter $m(r)$ remain unresolved.
- **Recorder and control arms.** A dual-recorder comparison that logs the same trajectory through active-block and full-canvas/passive instruments is GPU-gated future work; no cross-recorder result is claimed. A fresh in-session L32/NFE48 run is also GPU-gated: the present control reuses a pre-existing contiguous pool, has the lowest positivity (0.8805), and its percentile stability band touches zero.
- **Content scope.** The content association comes from one L32/NFE32 bed and survives only through moderate null granularity (it is not established at the finest registered BIN32 granularity). Its behavior on the L16/L64 beds is not measured.

5 RELATED WORK

Foundational discrete-diffusion work supplies modeling background (Austin et al., 2021; Zheng et al., 2024; Lou et al., 2024; Shi et al., 2024). The nearest verified architectural source is Arriola et al. (2025), which defines block diffusion as autoregressive across blocks with discrete denoising inside each block. It establishes the blockwise decoding context, but does not study active-recorder identity-lock times or the finite-grid selector used here.

Two checked measurement sources motivate only a narrow confound comparison. Sahoo et al. (2026) shows that apparent linear-probe separation can be explained by task-format confounding after residualization; it does not analyze diffusion recorder windows. Feng et al. (2024) studies propositional probes for latent world states and motivates distinguishing an internal-state measurement from model outputs; it does not perturb an active-block observation window. Table 12 records these verified boundaries. Within this checked set, the present paper examines a different measurement object. This is not a priority or field-wide absence claim: the bibliography-wide metadata and sentence-level support audit remains incomplete. The post-treatment amendment, strictly- L -supported boundary, and tested-family-only DFT check remain explicit.

Reproducibility statement. Reproducing the reported analyses requires the listed event streams, analysis scripts, and private model checkpoint. The audited package does not include them, so no numeric rebuild was run from that package.

6 CONCLUSION

In this recorder-relative evidence audit, the source-reported complete rows for two main arms and one reused L32 development control are consistent with the registered selector returning each configured block length. Bootstrap summaries describe sample stability, while the rank-sum ordering is an alternative statistic on those same profiles rather than independent calibrated inference. The L24/L48 records remain unauditable side observations, not headline confirmation, because their complete candidate vectors are unavailable; L24 is marginal under its reported conditional diagnostic and L48 is single-locus. The reporting recommendation is to state what set composition makes reachable before data and then separately state what measured magnitudes show. The descriptive result is limited to the active recorder under the composite partition-plus-required-allocation intervention and supports no model, sampler, recorder-causal, allocation-causal, cross-bed, or universal mechanism.

REFERENCES

Marianne Arriola, Aaron Kerem Gokaslan, Justin T. Chiu, Zhihan Yang, Zhixuan Qi, Jiaqi Han, Subham Sekhar Sahoo, and Volodymyr Kuleshov. Block diffusion: Interpolating between autoregressive and diffusion language models. In *International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=tyEyYT267x>.

- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*, volume 34, pp. 17981–17993, 2021. URL <https://arxiv.org/abs/2107.03006>.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. URL <https://arxiv.org/abs/2110.14168>.
- Anthony C. Davison and David V. Hinkley. *Bootstrap Methods and their Application*. Cambridge University Press, Cambridge, UK, 1997.
- Jiahai Feng, Stuart Russell, and Jacob Steinhardt. Monitoring latent world states in language models with propositional probes. In *Advances in Neural Information Processing Systems*, 2024. URL <https://arxiv.org/abs/2406.19501>.
- Ronald A. Fisher. Tests of significance in harmonic analysis. *Proceedings of the Royal Society of London. Series A*, 125(796):54–59, 1929. doi: 10.1098/rspa.1929.0151.
- Marjan Ghazvininejad, Omer Levy, Yinhan Liu, and Luke Zettlemoyer. Mask-predict: Parallel decoding of conditional masked language models. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pp. 6111–6120. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1633. URL <https://aclanthology.org/D19-1633/>.
- Jinsong Li, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Jiaqi Wang, and Dahua Lin. Beyond fixed: Training-free variable-length denoising for diffusion large language models. *arXiv preprint arXiv:2508.00819*, 2025. URL <https://arxiv.org/abs/2508.00819>.
- Aaron Lou, Chenlin Meng, and Stefano Ermon. Discrete diffusion modeling by estimating the ratios of the data distribution. In *International Conference on Machine Learning*, 2024. URL <https://arxiv.org/abs/2310.16834>.
- Thomas E. Nichols and Andrew P. Holmes. Nonparametric permutation tests for functional neuroimaging: A primer with examples. *Human Brain Mapping*, 15(1):1–25, 2002. doi: 10.1002/hbm.1058.
- Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Large language diffusion models. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL <https://arxiv.org/abs/2502.09992>.
- Shen Nie, Qiyang Min, Shaoxuan Xu, Zihao Huang, Yuxuan Song, Yong Shan, Yankai Lin, Wayne Xin Zhao, Chongxuan Li, and Ji-Rong Wen. Improved large language diffusion models. *arXiv preprint arXiv:2606.25331*, 2026. URL <https://arxiv.org/abs/2606.25331>.
- Subramanyam Sahoo, Vinija Jain, Aman Chadha, and Divya Chaudhary. Linear probes detect task format, not reasoning mode in language model hidden states. In *6th Workshop on Trustworthy NLP, ACL*, 2026. URL <https://arxiv.org/abs/2606.02907>.
- Jiaxin Shi, Kehang Han, Zhe Wang, Arnaud Doucet, and Michalis K. Titsias. Simplified and generalized masked diffusion for discrete data. *arXiv preprint arXiv:2406.04329*, 2024. URL <https://arxiv.org/abs/2406.04329>.
- Bian Sun, Kevin Zhai, Mubarak Shah, and Zhenyi Wang. Dystruct: Dynamically structured diffusion language model decoding via bayesian inference. *arXiv preprint arXiv:2605.09820*, 2026. URL <https://arxiv.org/abs/2605.09820>.
- Xinyun Wang, Min Zhang, Sen Cui, Zhikang Chen, Bo Jiang, Kun Kuang, and Mingbao Lin. Reversible diffusion decoding for diffusion language models. *arXiv preprint arXiv:2602.00150*, 2026. URL <https://arxiv.org/abs/2602.00150>.

Lin Zheng, Jianbo Yuan, Lei Yu, and Lingpeng Kong. A reparameterized discrete diffusion model for text generation. In *Proceedings of the First Conference on Language Modeling*, 2024. URL <https://arxiv.org/abs/2302.05737>.

A PROTOCOL CHRONOLOGY, MEASUREMENT CANON, AND SENSITIVITIES

A.1 EXPLORATORY SPECTRAL RESIDUAL CHECK

For a nested candidate such as $\mathcal{B}_{32} \subset \mathcal{B}_{16}$, a *negative* $\text{sep}(32)$ is not evidence of absence: the extremum comparison cannot recover every possible residual shape. An *exploratory* (post-data, not pre-registered) check therefore asks whether a *non-nesting* estimator recovers a residual source period that sep misses. We use a DFT periodogram: for each candidate P , $G(P) = \|\sum_r d(r)e^{-2\pi ir/P}\|/\|d\|$ is the normalized spectral magnitude of the adjacent-difference profile at frequency $1/P$, with a permutation null (shuffle $d(r)$ positions; 4,000 replicates) and family-wise q_{95} over all candidates. Harmonic testing follows the periodogram tradition (Fisher, 1929); our finite-grid threshold is a bespoke single-step maximum over candidates, using the standard permutation max-statistic construction (Nichols & Holmes, 2002). We also run a direct two-sample test of the two $L=16$ -boundary subsets in Table 6 (the odd-16 chain vs. the would-be- $\mathcal{B}_{32}$ positions), a reachable test of residual 32-amplitude modulation.

Case	$G(8)$	$G(16)$	$G(24)$	$G(32)$	$G(48)$	$G(64)$	$q_{95}@32$
Synthetic, residual-32 present	3.58	3.58	—	4.86	—	—	1.74
Synthetic, no residual	4.01	4.01	—	0.25	—	—	1.75
Real L16 measured	3.79	3.70	0.15	0.23	0.36	0.18	2.15 (FW)

Supporting check	Statistic	Threshold / reference	Reading
Injected residual	Residual-32 amplitude 0.5 over comb-16 amplitude 3.0; $G(32)=4.86$	$q_{95}=1.74$	fires
No-residual synthetic	$G(32)=0.25$	$q_{95}=1.75$	null
Real L16 remaining grid	$G(24)=0.15$, $G(48)=0.36$, $G(64)=0.18$	per-candidate $q_{95} \approx 1.71\text{--}1.78$	no residual candidate
Comb harmonic	$G(8)/G(16)=1.000$ (pure comb), 1.024 (real L16)	1.0 reference ratio	recorder-comb harmonic
Power sweep	Amplitude 0.15: no fire; 0.20: fire; grid step 0.05 (5.0%–6.7% of comb amplitude 3.0)	q_{95} above	transition bracketed
Odd-16 vs. would-be- $\mathcal{B}_{32}$	Mean $ \Delta d =0.036$; permutation $p=1.0$	$\alpha = 0.05$	no amplitude split

Table 7: Spectral (DFT) periodogram of the adjacent-difference profile $d(r)$ — an *exploratory*, non-nesting check (Section A.1). $G(P)$ is the normalized spectral magnitude at frequency $1/P$; “fires” means the 32-component exceeds its permutation q_{95} (4,000 replicates). The upper panel completes the synthetic diagnostic grid and reports every real candidate; the lower panel separates the supporting power, harmonic, and two-sample checks from their interpretation. The synthetic residual fires, whereas the comb-only synthetic and real L16 profile are null at the target period. On real L16, no candidate in the tested grid is significant; only the recorder-comb harmonics fire. The power sweep brackets sensitivity for the tested sinusoidal family only, and the direct boundary-subset comparison finds no split. Provenance is in Table 9.

Development control. On the recorder-consistent synthetic grid, with a clean sinusoidal residual-32 component overlaid on the $L=16$ comb, sep returns $P^*=16$ (blind) while the spectral 32-component fires; on the no-residual synthetic comb, both are silent (Table 7). This is a positive control for the tested sinusoidal family, not validation against arbitrary residual shapes.

Real L16 arm. The tested candidate grid does not exceed its family-wise threshold at $P = 32$, and the boundary-subset test finds no amplitude split. The registered power sweep brackets the

tested-family transition and records its amplitude range in Table 7. Hence the measured arm does not trigger for this tested sinusoidal family and amplitude range; arbitrary residual shapes are not ruled out (Table 9).

A.2 REGISTRATION CHRONOLOGY

Opaque stage	Relation to data	Source-reported event
R0	Pre-main-data	Original registration preceded the main-arm execution.
R1	Pre-main-data	A development amendment replaced a failed signed mean-difference score with the min-max estimator and recorded the selection-and-stability criterion.
R2	Main-data	The two main arms were executed.
R3	Post-main-data	A scope amendment withdrew the model-intrinsic branch.
R4	Pre-probe-data	A separate non-nested canvas-96 protocol and its amendments preceded the two sequential probes.

Table 8: Opaque registration sequence (SR/CV). Raw timestamps and searchable internal identifiers are omitted from blind prose. The underlying registrations, trusted timestamps, and diffs are absent from the audited package, so the sequence is source-reported and cannot be audited. It documents a post-main-data scope change, not a pre-main-data mechanism verdict.

A.3 OPAQUE EVIDENCE INDEX

Evidence ID	Role	Accessible-package status
A31-E1	Two main-arm and reused-control selector profiles	SR/CV; numeric targets and programs absent
A31-E2	Paired/marginal bootstrap and stability summaries	SR/CV; pairing manifest and replicate indicators absent
A31-E3	Sequential L24/L48 probes	SR/CV; complete candidate and rank-sum vectors absent
A31-E4	Conditional margin, rank-transform, and spectral checks	SR/CV; result arrays and programs absent
A31-E5	Separate content-association bed	SR/CV; events and analysis programs absent
A31-R1	Registration and amendment sequence	SR/CV; timestamped records and diffs absent
A31-V1	Figure 2	Raster present; source data and generator absent

Table 9: Blind-safe claim-to-evidence index. Opaque paper-local IDs replace raw workspace paths, timestamp strings, and digest or commit prefixes. The private crosswalk is unavailable in the audited package; this table is an availability disclosure, not evidence that the source-reported numbers were reproduced.

A.4 MEASUREMENT CANON

For blockwise decoding observed through an active-block recorder, a position-resolved analysis should:

1. record block length, per-block steps, and generated-canvas coordinates;
2. calibrate within-block windows to the generating run rather than a fixed model constant;
3. report reachability twice: *a priori*, what estimator algebra and boundary-set composition imply; *from events*, what the measured magnitude ordering implies. In particular, $\mathcal{B}_{32} \subset \mathcal{B}_{16}$ does not force $\text{sep}(32) \leq 0$;

4. perturb the schedule when attributing a position-resolved diagnostic;
5. test content claims against a structure-preserving null and report its granularity; and
6. mark cross-bed relationships unmeasured until the corresponding intervention is run.

This is a measurement canon, not a mechanism claim. The registered estimator returns $P^* = L$ on the displayed *strictly L-supported* profiles: non-degenerate jumps occur on all L boundaries and no L -multiple's exclusive-boundary residual outcores the L -boundary minimum. It is not a guarantee for arbitrary residual profiles, and identifying commit internals would require instrumentation of sampler-state transfer.

A.5 SELECTIVITY DISCLOSURE (PERMUTATION-GATE SIZE AUDIT)

Unlocated audit grid. A source summary reports that an original gate flagged 10/25 wrong candidates across the five evidence strata (SR/CV). The exact per-arm audit candidate lists and gate outputs are absent. In particular, each displayed canvas-256 selector grid has four candidates and therefore only three non- L candidates, so the reported denominator of five per arm cannot be derived from that grid. We do not guess an alternative grid, reproduce the per-arm fractions, or compare the aggregate with the displayed selector grid. The 10/25 summary is retained only as an unauditable source report from an unlocated audit grid whose relation to Table 1 cannot be verified.

Conditional margins and cross-condition red-team. Table 5 reports the separately supplied top-minus-second separation margins for the main arms and reused control. Table 4 keeps the L24/L48 limitations as side observations. The source also describes a global cross-block shuffle as a structureless reference, but its complete output bundle is absent. These construction-specific, unadjusted diagnostics do not calibrate the complete selector and are not per-arm false-positive rates.

Reporting rule. Any threshold-type permutation gate used here for selectivity reports three quantities together: the wrong-candidate hit rate, a known-clean-arm or explicit cross-condition check, and the discrete exact p . Because the source-reported 10/25 denominator cannot be mapped to a disclosed candidate list, no selectivity conclusion in this evidence audit rests on that aggregate.

A.6 THREE-PART EVIDENCE DECOMPOSITION

For auditability, we separate the recorder-relative object $d(r)$, the pre-treatment min-max decision procedure, and the scoped empirical finding that P^* follows L for the two main arms relative to a reused development control. The shared-cluster bootstrap probes sample stability, and the same-profile rank-sum alternative probes statistic sensitivity. The sequential non-nested canvas-96 probes remain unauditable side observations because their candidate vectors are incomplete. Table 9 maps each component to its availability status. This decomposition is a falsification aid, not three independent novelty claims or a mechanism attribution.

A.7 SUPPORTING CONTENT ASSOCIATION ON A SEPARATE BED

Estimand τ . On the separate L32/NFE32 bed, one unit is a decode-block with resolved positions $\mathcal{R}$; c_r is initial confidence and $o_r = r \bmod 32$ is block offset. The estimand is the median per-unit rank association after residualizing confidence on offset:

$$r_u = \frac{\sum_{p < q \in \mathcal{R}: \ell_p \neq \ell_q} \text{sgn}(\ell_q - \ell_p) \text{sgn}(\tilde{c}_p - \tilde{c}_q)}{\#\{p < q \in \mathcal{R} : \ell_p \neq \ell_q\}}, \quad \tilde{c}_r = \text{rank}(\text{rank}(c_r) - \hat{\beta} \text{rank}(o_r)), \quad (3)$$

Here the registered statistic fits $\hat{\beta}$ once on the observed data and shuffles the resulting $\tilde{c}$ within offset-rank strata. A sensitivity check instead refits $\hat{\beta}$ inside every permutation replicate. Earlier lock with higher residualized confidence contributes +1. Inference is rooted at decode clusters: the 500-replicate interval resamples decodes and carries their block units with multiplicity. The position-fake keeps the observed amplitude but reassigns it to positions, destroying the observed content-position pairing. This differently budgeted bed supplies supporting, not cross-block-length, evidence.

Quantity	Estimate / band	Null q_{95}	Null readout
Content-layer τ_{res}	0.2722 [0.2649, 0.2812]	BIN16 0.20235	0/2000 exceed
Refit- $\hat{\beta}$ sensitivity	0.2722	BIN8 0.13545	0/2000 exceed
Same-amplitude position fake	max. margin +0.00979	observed margin +0.12837	below observed

Table 10: Content-versus-structure result on the separate $L = 32$, NFE-32 bed. The statistic τ is the median decode-block rank association between earlier lock and higher confidence after offset residualization. BIN16 is the load-bearing conservative, non-degenerate null; the per-replicate nuisance refit is a separate BIN8 sensitivity and replays bit-exactly. Table 11 reports the full granularity ladder and three-bucket one-flip classification. This supporting analysis is not a cross-block-length experiment.

The full-granularity ladder in Table 11 is essential. The observed association clears every non-degenerate restricted null through BIN16, but matches the zero-power point mass at BIN32. Thus content association is established at moderate granularity, never at the finest registered granularity; the construction-specific BIN32 line is not interpreted as a lower bound. Its behavior under $L \in \{16, 64\}$ is unmeasured, and the two analyses are not claimed to be mutually calibrated.

A.8 CONTENT ACCOUNTING AND NULL GRANULARITY

Accounting / sensitivity check	Recorded value	Interpretation
Content-resolved block-units	6,178 of 6,274 candidates	Terminal-resolved units excluded
Resolved comparable pairs	1,571,854	Both positions resolve
Within-unit support	Median 4 distinct locks; median resolved-pair fraction 0.575	Unit-level order is observed
Resampling	$n_{\text{boot}}=500$; $R_{\text{null}}=2000$	Decode-cluster interval and restricted null
K-ladder restricted-null q_{95}	$K=1$: 0.00486; BIN4 0.09276; BIN8 0.14384; BIN16 0.20235; BIN32 0.27221 (point mass)	With $\tau_{\text{res}}=0.2722$, margins at $K=1$, BIN4, BIN8, and BIN16 are 0.26735; 0.17945; 0.12837; 0.06986; BIN32 is zero-power by construction and carries no lower-bound reading
Nuisance- $\hat{\beta}$ sensitivity	Refit inside each BIN8 replicate: $q_{95}=0.13545$; fixed-fit BIN8 $q_{95}=0.14384$	0/2000 exceedances; bit-exact replay. BIN16 remains the load-bearing conservative non-degenerate arm
One-flip sensitivity ($1/n_{\text{selected}}$)	$n_{\text{selected}}=6178$; all 6178 residual-sign flips give $\Delta=0$	DEGENERATE, by construction; no pass claim. The null-specification ladder, not this identity arm, carries the sensitivity conclusion

Table 11: Unit accounting and restricted-null sensitivity for Table 10. A block-unit is rooted in its decode cluster; decodes are resampled with multiplicity. The registered line fits the nuisance slope once; the displayed sensitivity refits it inside each replicate. One-flip granularity is $1/n_{\text{selected}}$, not permutation spacing $1/R$. Under the three-bucket contract, maximum single-point effect $\phi = 0$ is DEGENERATE (by construction) and carries no pass/fail claim; $0 < \phi \leq \text{margin}$ would pass and $\phi > \text{margin}$ would fail. BIN32 is likewise zero-power by construction. The full sweep therefore limits the positive finding to moderate granularity. Provenance is in Table 9.

A.9 CLOSEST-WORK SETUP COMPARISON

Verified source	Supported local use	Boundary relative to this paper
Arriola et al. (2025)	Block diffusion is autoregressive across blocks and uses discrete denoising within blocks.	Architectural context; no active-recorder lock-time or period-selector study.
Sahoo et al. (2026)	Task format can confound a linear probe’s apparent reasoning-mode separation after residualization.	Instrumentation-confound caution; no diffusion recorder-window perturbation.
Feng et al. (2024)	Propositional probes monitor latent world states and distinguish internal measurements from outputs.	Measurement-layer context; no active-block observation intervention.

Table 12: Narrow closest-work matrix restricted to the three sources with primary-source support verified for these local sentences. The rows do not establish priority or a field-wide absence; the bibliography-wide audit remains incomplete.

B REPRODUCIBILITY AND ARTIFACT MAP

Execution cost, shard completion, and event-row validation are reported in Table 13, separately from provenance, so that spend is never conflated with a completeness check. Table 9 uses opaque paper-local evidence IDs. The audited package contains the manuscript, bibliography, and two figures, but not the event, result, program, registration, or evidence bundle; the table is therefore an availability index rather than a self-contained artifact release.

Reported determinism claim. The source report states that decoding uses temperature 0 with *remasking=low_confidence* and that token selection is deterministic conditional on the checkpoint, seed, question assignment, canvas, and NFE schedule. It consequently describes $m(r)$, $d(r)$, $\text{sep}(P)$, and spectral values as deterministic properties of a recorded profile, while bootstrap/permutation summaries carry resampling randomness. The checkpoint fingerprint, exact inputs, implementation, and event schema are absent, so this claim cannot be verified here; cross-seed variation is not claimed.

Reported environment. The source records Python 3.11.13, NumPy 2.4.6, pandas 2.3.1, PyArrow 21.0.0, PyTorch 2.7.1; the model is a locally registered LLaDA-8B-Instruct-scale masked dLLM, blockwise decoded over GSM8K (Cobbe et al. 2021), a fixed 300-question canvas at 256 positions and 48 total NFE per arm. The checkpoint is a private local artifact (registry per the project evaluation ledger) and is *not* byte-identical to the public LLaDA-8B-Instruct release. No checkpoint fingerprint or registry record is available, so release equivalence and the reported scale cannot be checked.

Reported cross-environment replay. The source report describes a bit-for-bit replay of the three canvas-256 arm vectors in a second software environment (Python 3.12.3, NumPy 2.5.2, pandas 3.0.5). The raw events, both program versions, outputs, and full digests are absent. This audit did not rerun that analysis and cannot verify the replay; it is not presented as new or independent evidence.

Reproduction boundary. The source reports a fixed seed and a contiguous 300-question development subset, but their searchable internal identifiers are omitted from blind prose and the private crosswalk is unavailable. The audited package does not ship event streams, executable programs, registrations, machine-readable results, or the private checkpoint and hence does not support a numeric rebuild. No numeric rebuild was executed for this audit. A reviewer-safe supplement remains an unresolved external requirement.

Execution accounting	Recorded value
Hardware	eight GPUs
Timing	330 running s = 0.73 GPU-h; 130 queueing s reported separately and excluded
Shard completion	$L = 16$: 4/4; $L = 64$: 4/4
Event-row check	$L = 16$: 153,600 = 300×512 ; $L = 64$: 499,200 = 300×1664 ; $L = 32$ control: 268,800 = 300×896 rows for the reused contiguous subset of the frozen NFE-48 pool

Table 13: Source-reported execution cost and completion checks (SR/CV) for the two main arms and reused control subset. These values cannot be checked from the audited package; the rows describe the inherited report rather than a run made for this audit.

AI-assisted preparation disclosure. The source report labels qualitative Figure 1 as image-generated, but no model, prompt, or generation record is present in the audited package; only its non-evidentiary role can be checked. No image generation was used for this audit. Figure 2 is labeled data-derived by the source, but its data and generator are absent. An AI assistant supported language editing, LaTeX refactoring, and internal review. All empirical values remain explicitly SR/CV rather than being represented as AI-verified or rerun; the authors retain responsibility for scientific content.

C NOTATION AND TERMINOLOGY

Table 14 collects the estimator notation and verdict vocabulary; Table 15 collects the bed names, the treatment variable, and the content-layer terms. Each entry names the section that defines it.

Table 14: Estimator notation and legacy decision-table vocabulary; terms point to their operative definition and do not attribute mechanism.

Term	Meaning (defining location)
$\ell_{i,r}$	Identity-lock time: earliest recorded step after which decode i ’s token at relative position r never changes again (Eq. (1)).
$m(r), d(r)$	Per-position median lock time over decodes, and its adjacent difference $m(r+1) - m(r)$ (Section 2).
$\mathcal{B}_P, \mathcal{I}_P$	Candidate- P boundary set $\{r : r \bmod P = P - 1\}$ and its complement, the candidate interior (Section 2).
$\text{sep}(P), P^*$	Min-max period separation $\min_{\mathcal{B}_P} d - \max_{\mathcal{I}_P} d$, and its maximizing candidate (Eq. (2)).
SELECTED	$\text{sep}(P^*) > 0$, bootstrap modal period = P^* , mode fraction ≥ 0.95 (registered operational criterion, not a null-calibrated test; Section 2).
REFUTED / NOT-SUPPORTED	Measured $\text{sep}(P) < 0$ for a candidate that the decision table treated as refutable / for the originally registered intrinsic-32 conjunct, whose branch was withdrawn post-data (Table 6).
A-PRIORI layer	What follows from the estimator’s algebra alone, before data — e.g. boundary-set nesting (Section 2).
FROM-EVENTS layer	What follows only once measured magnitudes exist — e.g. the sign of $\text{sep}(32)$ (Section 2).
Strictly L -supported	Recorded profile with non-degenerate jumps on all L -block boundaries such that no L -multiple’s exclusive-boundary residual outcores L ’s boundary minimum (Section 3.1).
$G(P)$, perm. q_{95} , fam. q_{95}	Normalized DFT magnitude of $d(r)$ at frequency $1/P$; its per-candidate permutation 95th percentile; the family-wise version over all candidates (Section A.1).

Term	Meaning (defining location)
L16 arm, L64 arm	The two new matched treatment arms at block length 16 and 64, NFE 48 (Table 1).
L32/NFE48 control	Frozen matched positive control at block length 32, NFE 48: a registered contiguous development subset, $n=300$, with opaque internal identifiers (Sections 2 and 3.1).
L32/NFE32 content bed	Separate, differently budgeted bed used only for the content-association diagnostic (Section A.7).
Partition-plus-required allocation	The composite treatment: changing the block partition necessarily changes the per-block step allocation at fixed total NFE (Table 1).
Measurement-class label	PERIOD-FOLLOWS-BLOCK-LENGTH: a post-data label stating that the selected boundary period follows block length on the tested bed under the active-block recording window; not a mechanism attribution (Sections 1 and 3.1).
BIN8	Restricted null that permutes residualized confidence within 8 equally populated offset-rank bins, preserving offset structure to 4-position granularity (Sections 2 and A.7).
conf0	Position- r initial confidence at block-commit: the model’s confidence for r at the first recorded step of its block (Section 2).
Block-unit; resolved position	One decode-block of the content bed; a position whose lock time is defined and that never flips after lock. Terminal-resolved units, which present no within-block pairwise order, are excluded (Table 10).
τ, τ_{res}	Median over units of the per-unit rank association between lock time and confidence; the offset-residualized (content-layer) version (Eq. (3)).

Table 15: Bed, treatment, and content-layer terms; the matched setting and separate content bed remain distinct.